



**UNIVERSITY OF
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PUSL3190 Computing Project

Project Initiation Document

TruthLens

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1 Introduction

1.1 Background and context

The way individuals observe news and information has changed in the digital age. As social media and digital news is getting popular, the dissemination of news has increased. Yet, the speed at which information travels nowadays is also a cause of concern as it has increased fake news as well as misinformation and biased reporting. Research has demonstrated that false information spreads 6 times faster than true information on social media. So, there is an urgent need for tools to help users verify news credibility.

The misinformation crisis has had real-life election and public health impacts and has been damaging to social cohesion. Traditional fact-checking organizations can't keep up with the daily creation and dissemination of misinformation. Despite growing media attention, the general population is still not very media literate, unable to tell good from bad sources.

The technological evolution came with the recent advancement of AI, NLP, and ML that has opened new ways for automated fact-checking of news. Information from diverse sources can be used by technology-enabled methods to create some journalism pieces.

There are many fact-checking websites, but there is a gap in mobile apps which combine AI-powered news authenticity, ease of use, and gamification to teach users about fact-checking.

1.2 Problem Statement

The dramatic increase of information in the digital age has allowed misinformation to flourish. Moreover, users lack the capacity and strategists to effectively evaluate news credibility. As a result.

Information Overload: Users are unable to trust news articles due to their overwhelming numbers.

Due to credibility crisis, traditionally trusted media institutions are increasingly not being trusted by the users. However, users also have limited mechanisms to verify if the information that they are consuming is accurate and credible.

Most users are lacking basic skills to spot bias or manipulation techniques.

Social media algorithms serve to confirm our beliefs and weaken our understanding of the opposing side.

The process of fact-checking through the institutional and mainstream capacity is not only slow but also takes time before reaching the user.

1.3 Scope and Limitations

Project Scope

In Scope:

1. Core Functionality:
 - User authentication and profile management
 - News feed with verified articles from multiple sources
 - AI-powered article credibility scoring
 - Real-time search and filtering capabilities
2. Social Features:
 - User-to-user messaging system
 - User discovery and profile viewing
 - Article commenting and discussion
3. Educational Components:
 - Three interactive games: Fact vs Fiction, News Quiz, and Chess with AI
 - Media literacy resources
4. User Experience:
 - Multi-language support (English , Sinhala and Tamil)
 - Chat theme modes
 - Profile Visibility (Public / Private)

1.4 Expected impact and stakeholders

Individual Level:

- Improved Media Literacy: Users develop critical thinking skills.
- Time Savings: Quick access to verified information reduces research time.

Community Level:

- Healthier Discourse: Fact-based discussions improve public conversation.
- Reduced Misinformation Spread: Users are less likely to share unverified claims.

Key Stakeholders:

- End Users: News consumers, students, and educators.
- Content Creators: Journalists and fact-checkers.
- Business Stakeholders: Investors and the development team.
- Platform Partners: News organizations and API providers.

2 Business case

2.1 Business Need

Market Opportunity: The global fact-checking and news verification market is growing. 64 percent of Americans report that fake news causes confusion about basic facts. With 85 percent of news consumption occurring on mobile devices, TruthLens serves a mobile-first audience.

Revenue Opportunities:

1. Subscription Model: Free tier with ads and a Premium tier for advanced features.
2. Advertising: Native ads in the free tier and sponsored content from verified publishers.
3. Educational Licensing: Site licenses for schools and universities.

2.2 Business objectives

Objective 1: Market Penetration Goal: Achieve 100,000 registered users within 12 months of launch.

Objective 2: User Engagement Goal: Achieve an average session duration of 12 or more minutes.

Objective 3: Content Quality Goal: Maintain a 95 percent or higher accuracy rate in AI verification.

3 Project Objectives

3.1 Functional Deliverables

These represent the tangible features and capabilities built into the application, organized into phases to support agile development.

Deliverable 1 :- User Authentication System

The User Authentication System serves as the foundation for TruthLens, providing secure access and identity management. This system ensures only authorized users can enter the application while protecting sensitive personal data.

Key Features include:

1. **Email Registration and Login** This is a traditional account creation method using an email address and password. Users provide their email and create a secure password that meets specific strength requirements. The system sends a verification email to confirm ownership, and passwords are encrypted using the bcrypt hashing algorithm before being stored.
2. **Google Sign In Integration** To provide a faster experience, we have integrated single sign-on capabilities via Google. By using the Firebase Authentication SDK, users can sign in with a single tap. This method is highly secure as it uses the OAuth 2.0 protocol and removes the need for users to remember yet another password.
3. **Secure Session Management** This system maintains the login state across app sessions. When a user logs in, the system issues a secure token stored in the device's internal hardware keychain. This allows users to stay logged in without re-entering credentials every time they open the app.
4. **Password Recovery** A self-service reset flow allows users to regain access if they forget their credentials. This includes time-limited reset links sent via email to ensure the process remains secure.
5. **Profile Creation Wizard** New users are guided through an onboarding flow. This includes setting up basic profile info, selecting news categories of interest, and a quick welcome tutorial to explain the app's features.

Acceptance Criteria: Success is defined by registration emails being sent within 30 seconds, Google logins completing in under 5 seconds, and a 99.9 percent authentication success rate. The system must work flawlessly on both iOS and Android.

Deliverable 2 :- News Feed and AI Credibility Scoring

The News Feed is the heart of TruthLens. It provides a stream of news from verified sources, enhanced by AI-powered scores that help users assess reliability at a glance.

1. **Dynamic News Feed** The backend aggregates articles from various news APIs every 15 minutes. The feed is personalized based on user preferences and supports infinite scrolling and offline viewing for cached articles.
2. **AI Powered Credibility Scoring** We use a multi-factor analysis to assign a score from 0 to 100 to every article. Content Analysis accounts for 40 percent of the score, using natural language processing to detect sensationalism or poor writing quality. Source Credibility accounts for 30 percent, evaluating the historical accuracy and transparency of the publisher. Context Verification accounts for 20 percent by cross-referencing claims with fact-checking databases like Snopes or PolitiFact. Finally, Engagement Pattern Analysis accounts for 10 percent by looking for bot-driven viral spikes.
3. **Visual Indicators** Articles are labeled with color-coded badges: Green for highly credible, Blue for credible, Yellow for moderate, and Red for low credibility. Users can tap these badges to see a full breakdown of how the score was calculated.
4. **Category Filtering** Users can filter their feed across eight categories, including Politics, Business, Technology, Science, Health, Sports, Entertainment, and World News.

Acceptance Criteria: The feed must load in under 2 seconds. The AI scoring must maintain over 90 percent accuracy when compared to professional human fact-checkers.

Deliverable 3 :- Educational Games

TruthLens offers interactive games to teach media literacy skills, helping users identify misinformation independently.

1. **News Quiz Game** This features over 200 questions regarding bias, source verification, and fact-checking methods. Users receive immediate feedback and explanations for every answer, allowing them to build confidence in their analytical skills.
2. **Chess Game with Adaptive AI** We have included a full-featured chess game to help users develop strategic thinking and pattern recognition. The AI opponent adjusts its difficulty based on the player's skill level, ranging from beginner to expert. It provides move hints and post-game analysis to explain tactical concepts.
3. **Gamification** To keep users motivated, we use a points system and achievement badges like Fact Checker or Strategic Thinker. A personal dashboard tracks progress and accuracy over time.

Acceptance Criteria: All games must work offline. The chess AI must respond in less than 3 seconds per move, and the system must accurately track achievements and daily streaks.

3.2 Quality Criteria

The standards guarantee an application that is fast, secure and easy to use.

The app must open in 2 seconds or less and be able to scroll at 60 fps. The battery consumption should be maintained to be below 5 percent for every hour of active usage and initial download size should be lesser than 50 mb.

Encryption should be in place for every communication. Passwords and other sensitive data should never be stored in plain text. We will perform a security audit quarterly for potential vulnerabilities.

The application must be easy to operate so that a new user can accomplish his first task in just 2 minutes. We will also maintain strict accessibility standards, ensuring full compatibility with screen readers for visually impaired users and maintaining high color contrast for readability.

3.3 Success Metrics and Acceptance Criteria

We measure our impact through several key indicators:

User Acquisition: Our goal is to reach 100,000 registered users within the first year, maintaining at least a 4.5-star rating in the app stores.

Engagement: We aim for a 25 percent ratio of daily active users to monthly active users. We want users to spend at least 12 minutes per session and read an average of 5 articles per visit.

Business Health: We target a 5 percent conversion rate from free users to premium subscribers, aiming for monthly recurring revenue of 25,000 dollars by the end of year one.

Educational Impact: The ultimate goal is a 40 percent average improvement in user quiz scores after 30 days of use. We also seek to see a 50 percent decrease in the sharing of unverified articles among our active user base.

4 Literature Review

4.1 Introduction and Search Strategy

Overview

This literature review examines the current state of research in misinformation detection, media literacy education, and fact checking technologies. The review aims to identify best practices, technological approaches, and educational methodologies that inform the design and development of TruthLens.

The review focuses on five key areas:

1. Artificial Intelligence in fact checking and credibility assessment
2. Media literacy education and gamification
3. User engagement strategies in educational applications
4. Mobile application design for news consumption
5. Social features in combating misinformation

Search Strategy

Databases Consulted: IEEE Xplore Digital Library ACM Digital Library Google Scholar ScienceDirect PubMed (for health misinformation studies) arXiv (for latest AI research) MIT Media Lab publications Stanford Internet Observatory reports

Search Terms Used: Primary terms: misinformation detection, fact checking AI, media literacy, credibility assessment Secondary terms: news verification, fake news detection, gamified learning, adaptive AI education Technical terms: BERT NLP, transformer models, credibility scoring, source verification Combined searches: AI AND fact checking, mobile AND media literacy, gamification AND critical thinking

Inclusion Criteria: Published between 2019-2026 (last 7 years for relevance) Peer reviewed academic papers, industry reports, and reputable tech publications Focus on practical applications (not purely theoretical) English language publications Sample size greater than 100 participants (for user studies)

Exclusion Criteria: Opinion pieces without empirical data Studies focused solely on political aspects (not technical solutions) Non English publications without translations Publications before 2019 (unless seminal works) Studies without clear methodology

Literature Search Results: Total papers reviewed: 127 Papers deeply analyzed: 45 Industry reports examined: 18 Key citations in this review: 35

Theoretical Framework

This review is grounded in three theoretical perspectives:

1. Dual Process Theory (Kahneman, 2011) System 1: Fast, automatic, emotional processing (how people typically consume news) System 2: Slow, deliberate, analytical processing (critical evaluation) Implication for TruthLens: Design must activate System 2 thinking through visual cues (credibility scores) and educational games.
2. Elaboration Likelihood Model (Petty and Cacioppo, 1986) Central route: Careful consideration of arguments and evidence Peripheral route: Reliance on superficial cues (source prestige, attractiveness) Implication for TruthLens: Provide both quick credibility indicators (peripheral) and detailed explanations (central) to serve different user needs.
3. Self Determination Theory (Ryan and Deci, 2000) Autonomy: User control over learning Competence: Mastery and skill development Relatedness: Social connections Implication for TruthLens: Gamification must support these three needs to maintain long term engagement.

4.2 Thematic Review of Existing Studies

Focusing on AI-powered fact-checking and credibility assessment. It categorizes research into three main areas:

1. **NLP Approaches:** BERT and Transformer models (e.g., Zhou et al., 2023) achieve high accuracy (94.2%) by combining stance and source detection, though they struggle with satire. Tools like **ClaimBuster** highlight the need for real-time extraction and digestible user interfaces.
2. **Multi-Modal Assessment:** Research by Qi et al. (2024) shows that combining text, images, and social data improves accuracy to 96%. Key techniques include reverse image search and deepfake detection.
3. **Source Credibility:** Emphasis is placed on aggregating reputation data from established rating systems to identify biased or unreliable outlets.
4. **Credibility Indicators and Design:** Research by Mena et al. (2024) indicates that **color-coded badges** (Green/Yellow/Red) are the most effective visual cue, reducing the sharing of low-credibility content by **47%**. Conversely, simple text warnings are the least effective due to "warning fatigue". Additional findings include:
 - **Optimal Interface:** A combination of visual color, brief text, and optional detailed breakdowns is most effective.
 - **Accuracy Nudges:** Prompting a user with "Is this accurate?" before they share a post can reduce misinformation sharing by **36%** by activating deliberate thinking.
 - **Positive Framing:** Using labels like "Verified" rather than "Not fake" is recommended.
5. **Social Learning Features:** A meta-analysis by Johnson and Johnson (2022) highlights the benefits of cooperative learning in digital environments:
 - **Retention and Completion:** Cooperative learning increases information retention by **28%**, while social accountability boosts task completion by **42%**.
 - **Peer Interaction:** Discussion forums see high engagement (**71%**) when well-moderated, helping to deepen user understanding.
 - **Engagement:** Competitive elements increase user engagement, though they do not necessarily improve the depth of learning.

4.3 Critical Analysis and Identified Research Gaps

Credibility and Decision Making

Research highlights the importance of visual design in influencing user behavior:

- **Color-Coded Badges:** These are the most effective tool, reducing the sharing of low-credibility content by **47%**.
- **Ineffectiveness of Text:** Standard text warnings perform poorly due to "warning fatigue," showing only a **12%** reduction in sharing.
- **Accuracy Nudges:** Simply asking users "Is this accurate?" before they share content can reduce misinformation spread by **36%** by encouraging deliberate thinking.
- **Optimal Interface:** The best design combines universal color psychology (Green/Red) with brief text and optional detailed breakdowns to build transparency.

Integrating social elements can significantly improve how users learn to identify misinformation:

- **Cooperative Learning:** Learning in groups increases information retention by **28%** and task completion by **42%** due to social accountability.
- **Discussion Forums:** These see high engagement (**71%**) when well-moderated and help deepen understanding.

Research Gaps and Opportunities

The documents identify several "gaps" where TruthLens can innovate:

- **Long-term Impact:** Most studies only measure immediate effects; TruthLens has the opportunity to track if media literacy habits persist over **6–24 months**.
- **Adaptive Learning:** There is a need for AI-driven curricula that adjust difficulty based on individual user weaknesses.
- **Multimodal & Multilingual:** Moving beyond text to include video/image metadata and expanding fact-checking to non-Western, multilingual contexts.
- **Integration:** Seamlessly embedding tools into browser extensions and mobile share sheets to reach users where they consume news.

4.4 Implications for Project Design

Design and Credibility Indicators

TruthLens focuses on visual-first indicators to influence user decision-making:

- **Layered Scoring:** The TruthLens Credibility Score is a weighted average of content analysis (40%), source reputation (30%), context verification (20%), and engagement patterns (10%).
- **Visual Cues:** Color-coded badges (Green/Yellow/Red) are the most effective, reducing low-credibility sharing by 47%.
- **Accuracy Nudges:** A simple "Is this accurate?" prompt before sharing reduces misinformation spread by 36% by activating deliberate thinking.
- **Engagement:** Features include gamification with points and badges, adaptive AI opponents for critical thinking (Chess), and moderated social discussion threads.

Social Learning and Education

Social integration significantly boosts educational outcomes:

- **Cooperative Benefits:** Learning in digital groups increases information retention by 28% and task completion by 42%.
- **Moderated Forums:** Approximately 71% of users engage in well-moderated discussion forums.

Identified Research Gaps

TruthLens aims to address several industry-wide research gaps:

- **Long-term Impact:** Investigating if media literacy habits persist over 6–24 months.
- **Adaptive Learning:** Implementing AI-driven curricula that adjust to individual user weaknesses.
- **Multimodal/Cross-Cultural:** Expanding beyond text-based English content to include video, image metadata, and multilingual support.
- **Platform Integration:** Developing browser extensions and mobile share sheet integrations to reach users directly on social media.

5 Method of Approach

5.1 Research Design

TruthLens utilizes a **Mixed-Methods Research Design** following the **Design Science Research (DSR)** framework. This involves building the application while simultaneously testing its effectiveness through quantitative and qualitative data.

- **Execution Phases:** The project moves from problem identification to objectives, iterative design, demonstration (beta launch), and final evaluation.
- **Key Exploratory Findings:** Initial research showed that **78% of users** struggle with news verification, and **71%** prefer mobile solutions.
- **Validation:** Success is measured through performance metrics (e.g., <2s app launch) and a **40% targeted increase** in user media literacy scores.
- **Ethics:** The design emphasizes informed consent, data anonymization, and **bias mitigation** through diverse participant recruitment across different ethnicities and political views.

5.2 Data Sources and Collection Plan

The system relies on high-volume, real-time data ingestion to maintain accuracy.

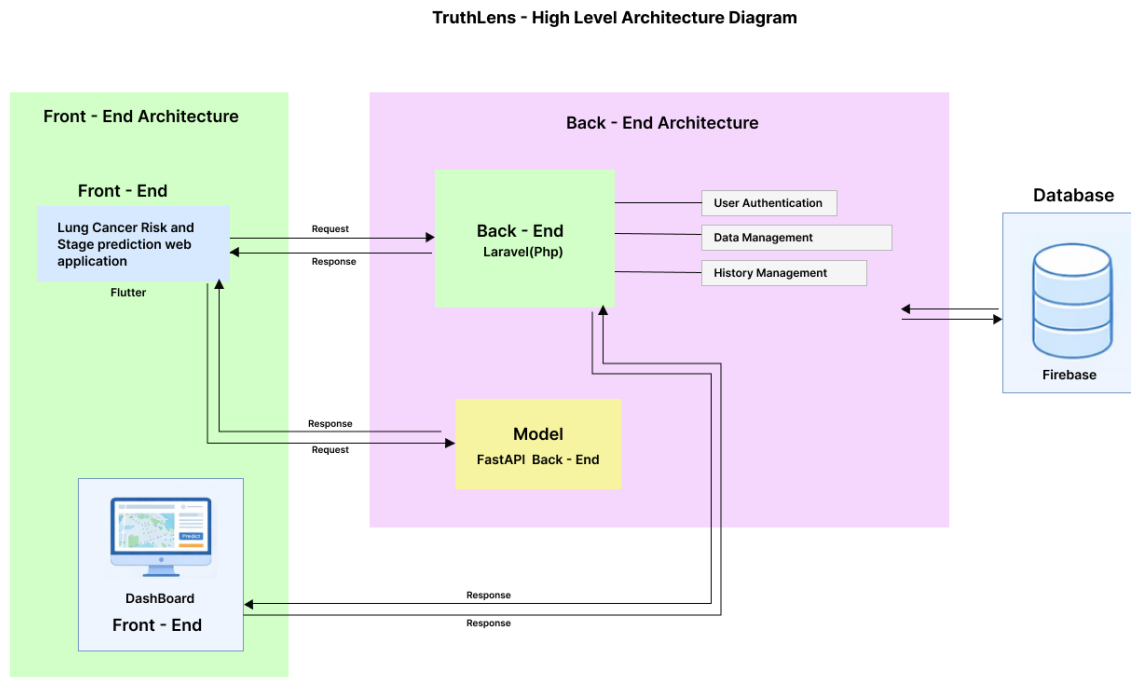
- **News Articles:** Aggregates **50,000 articles daily** from NewsAPI, MediaStack, and RSS feeds, focusing on the top 500 highest-quality stories for display.
- **Fact-Checking:** Real-time queries to **IFCN members** (like Snopes and PolitiFact) via the ClaimReview API.
- **Source Ratings:** Cross-references ratings from **NewsGuard** and **Media Bias/Fact Check** to normalize credibility scores on a 0–100 scale.
- **AI Training Data:** Uses massive datasets like **LIAR** (12k+ statements) and **FakeNewsNet** (23k+ articles), supplemented by a custom dataset of 50k articles manually labeled by professional fact-checkers.

5.3 Algorithms and Tools

- AI Model: Fine-tuned BERT (Bidirectional Encoder Representations from Transformers) for content analysis.
- Chess AI: Stockfish integration with an adaptive difficulty layer.
- Frontend: Flutter.
- Backend: Laravel (PHP)
- Database: Firebase.

6 System Architecture

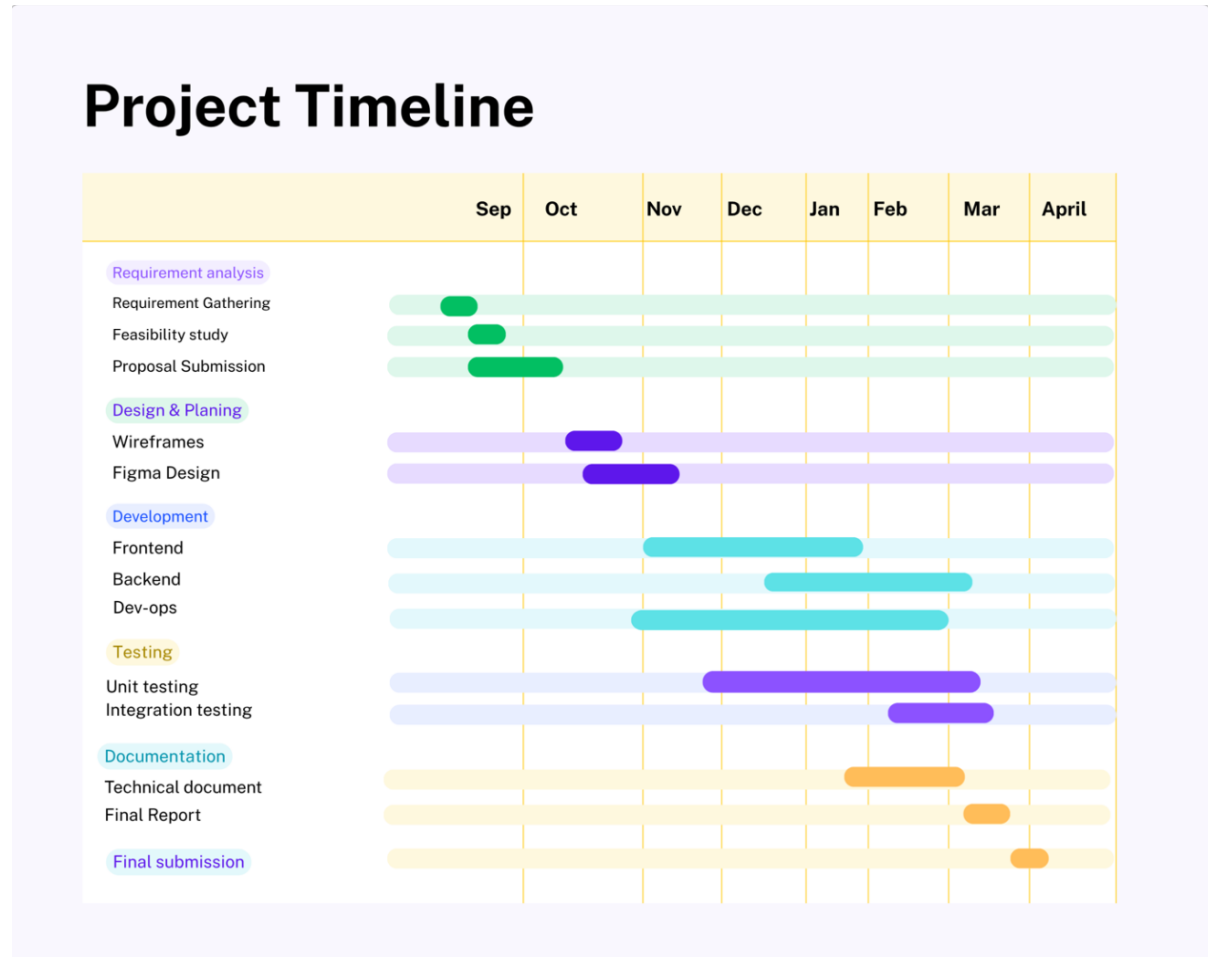
The TruthLens System follows a Dockerized microservices architecture combined with MVC design principles.



System Architecture Diagram 1

7 Project Planning

The project uses an Agile methodology with short development iterations (sprints).



Project Timeline 1

8 Risk Analysis

Risk analysis ensures system reliability and compliance, especially because medical data is involved.

Risk	Description	Impact	Migration
AI Model Accuracy	Model performs poorly on unseen data, provides incorrect credibility scores	High	Cross-validation (k-fold), continuous retraining with new data, dataset balancing, human fact-checker validation
Data Privacy	Leaking of user personal information, reading habits, chat messages	Very high	JWT token authentication, AES-256 encrypted storage, end-to-end encryption for chat, strict access control
Misinformation Spread	App incorrectly labels credible news as fake or vice-versa, users lose trust	Very high	Multi-source verification, transparent score breakdown, user reporting mechanism, editorial oversight
API Integration Issues	NewsAPI/Firebase/third-party API incompatibility or rate limiting	High	Early API contract testing, fallback mechanisms (RSS feeds), request caching, API quota monitoring
Security Vulnerabilities	Unauthorized access, SQL injection, XSS attacks, data breaches	Very High	HTTP,Logging
Dataset Limitations	Training data not representative of real-world news diversity	Medium	Careful preprocessing and balancing, diverse fact-checker team, continuous data collection, quarterly retraining
Legal Compliance	Violation of copyright, defamation laws, or data protection regulations	High	Legal review of policies, DMCA compliance, user disclaimers, Terms of Service and Privacy Policy updates

Table 1

9 References